

Compact Vision–Language Models for Cross-Crop Plant-Disease Diagnosis at the Edge

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Abstract

Cloud vision–language models diagnose plant disease well but bill per query and need connectivity, and a conventional classifier has no output unit for an unseen crop. We ask what lets a small, frozen vision–language model diagnose crops it was never trained on. On the SAGE dataset, four frozen compact contrastive encoders of 11.4–86.3 M parameters match leaf images to written disease descriptions, and we compare seven ways of authoring those descriptions across nested held-out sets of 16, 34 and 51 unseen classes. First, authoring outweighs model size: the best descriptions gain 10.3–16.2 points over a bare class name, against 4.0–5.1 points between the smallest and largest encoder, and authoring exceeds size at every configuration on both the full and a de-duplicated label set. Second, per-class sentences describing each disease as it appears in a photograph reach 30.9% top-1 on 51 unseen classes, 7.4 times the majority-class prior, and beat source-grounded descriptions by 7.0 points on all four encoders. Third, requiring a citable source costs no measurable accuracy, and re-embedding the source-grounded text as a sentence ensemble recovers only 4% of its deficit, so the gap lies in the text rather than in citation or embedding. The smallest encoder runs at 17.4 ms per image on a laptop CPU in 45.8 MB. For edge deployment, effort spent on descriptions returns more than effort spent on parameters.

1. Introduction

Vision–language models (VLMs) served from the cloud now diagnose plant disease with high accuracy in the wild, but they are metered per query and need a network connection. Both are a poor fit for smallholder and field use, which needs inference that is offline, cheap and fast. A second obstacle runs deeper. Models trained on one set of crops transfer poorly to others [3], and a conventional classifier cannot diagnose a crop it was not trained on at all: it has no output unit for the class. Diagnosing an unseen crop requires matching an image against a description of each disease, which is what an image–text-aligned model offers and a classifier does not.

This article asks what makes a *small* such model work. We freeze a compact Contrastive Language–Image Pretraining (CLIP) image encoder, precompute one text prototype per disease from a written description, and diagnose an unseen crop by returning the disease whose prototype is nearest to the image. Only the image encoder ships to the device. Two things can then be varied: the encoder, and the text. The field has concentrated on the first. We find the second matters more.

Contributions

1. **Descriptor authoring is the dominant lever, ahead of model size.** Across seven authoring strategies, three nested label spaces and five encoders, the best descriptions gain 10.3–16.2 points of unseen-crop

accuracy over a bare class name, while moving from an 11.4 M to an 86.3 M encoder gains 4.0–5.1. Authoring exceeds size at every configuration, on both the full label set and a de-duplicated one (Section 4.1).

2. **Photo-descriptive per-class sentences are the best authoring method tested.** Written in the manner of Customized Prompts via Language models (CuPL) [14], they reach 30.9% on 51 unseen classes, 7.4 times the majority-class prior, and lead every other strategy at the two larger label spaces on every deployable encoder (Section 4.1).
3. **A decomposition of what separates the methods.** Requiring each description to cite a retrievable source costs no measurable accuracy; how the prototype is built explains almost none of the remaining gap; the difference lies in the text itself (Section 4.2). Auditability is therefore available without an accuracy penalty, but citing sources does not by itself produce better descriptions.
4. **An edge benchmark** with measured CPU latency and INT8 footprint, and the finding that INT8 slows hybrid convolution–transformer encoders on x86 while accelerating the pure transformer (Section 4.4).

2. Related work

Agricultural VLMs SAGE [1] provides our data and motivates our descriptor design: its source-grounded schema, in which every symptom field carries a value, a source URL and a verbatim quote, and its reported 14–16 point gain from symptom knowledge. We are an edge counterpart to its cloud agent. Once a descriptor registry has been authored,

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inference costs nothing per image; the cost is one-time, in generation and verification. The agricultural foundation-model landscape is surveyed by Li et al. [8], whose framing of the open problem as adaptation and deployment cost rather than raw capability is the gap we address. AgriCLIP [12] adapts a CLIP backbone to agricultural imagery, and Lu and Chen [9] report that zero-shot VLMs still trail supervised models on plant disease by a wide margin, consistent with the modest absolute accuracies here and the reason we also report shortlists and abstention.

Classification by description Two methods ask a large language model (LLM) to describe each class and classify by matching images to those descriptions. Classification by description (DCLIP) [10] elicits short lists of visual features per class and scores a class by the mean similarity to its features. CuPL [14] elicits full sentences describing what the class looks like and averages their embeddings into one prototype. Both were developed on object-recognition benchmarks. We implement both and compare them with source-grounded, keyword and class-name descriptions on cross-crop plant-disease transfer, a setting in which neither has been evaluated.

Compact CLIPs CLIP [15] established image–text contrastive pretraining, and MobileCLIP [18], MobileCLIP2 [5] and TinyCLIP [19] compress it. We use their `open_clip` [7] releases as frozen backbones, with SigLIP 2 [17], whose sigmoid objective follows Zhai et al. [21], as a larger reference. Self-supervised encoders such as DINOv2 [13] learn strong visual features but have no text tower, so they cannot match an image to a written description; zero-shot diagnosis of an unseen crop requires an image–text-aligned model by construction.

Selective prediction Declining low-confidence inputs is selective prediction, formalised by El-Yaniv and Wiener [4] and extended to deep networks by Geifman and El-Yaniv [6]. Our margin-ranked gate is an instance of it, evaluated on the risk–coverage curve those works define. Deciding whether an input belongs to the trained label set at all is open-set recognition [16]; the seen/unseen split we assume given is the generalised zero-shot setting surveyed by Xian et al. [20], and we leave it unsolved (Section 6).

3. Materials and methods

3.1. Dataset and the nested design

We use SAGE [1], drawing images from its released dataset [2] (~839 K images over 335 crops and 1,251 disease classes, MIT licence) rather than legacy laboratory corpora such as PlantVillage [11], on which in-distribution accuracy has long been saturated. We filter to our crops with content-hash de-duplication and store images at a longest edge of 288 pixels. Classes are capped at 600 images for held-out crops and 1,000 for seen crops, and need at least 25 images to enter; images are taken in sorted filename order up to the cap. Filenames are content hashes, so the cap acts as

a deterministic random sample and the subset is exactly rebuildable. The caps also bound class imbalance: they keep the majority-class baseline at 4.2% on the largest held-out configuration, so a frequency prior cannot masquerade as transfer. The resulting subset is 84,123 images over 217 classes and 18 crops.

We pin the dataset to revision `bc9bd2899f` (7 May 2026). SAGE was re-released on 24 August 2026 with re-canonicalised names, and that release drops Cotton; every number here refers to the pinned revision, which the released fetch code also pins.

We define three **nested** configurations, in which every crop held out in one is also held out in the next. Configuration A holds out Coffee, Orange and Peach (16 classes, 4,050 images); B adds Cotton, Wheat and Bean (34 classes, 9,787 images); C adds Banana and Cucumber (51 classes, 14,204 images). Held-out crops are never trained on at any stage. Nesting prevents a crop from disappearing between configurations, but each step also adds new crops, so movement from A to C mixes a widening label space with a changing crop mix. We therefore read A, B and C as three operating points rather than as a controlled scaling curve.

Label noise and the de-duplicated set The held-out label set contains five alias pairs, in which SAGE names one disease twice (for example Orange/Canker and Orange/Citrus_Canker), and four labels for which no symptom description can exist (three wheat resistance ratings and one non-disease label). No description can separate a synonymous pair, so each carries an accuracy ceiling near 50%. We therefore report every result on two label sets: the full set as released, and a de-duplicated set that merges each alias pair and drops the four non-disease labels, leaving 13, 26 and 42 classes. Every conclusion in this article holds on both.

3.2. System

One frozen image–text encoder is paired with a zero-shot descriptor head that returns the class whose text prototype has the highest cosine similarity to the image embedding, and a gate that abstains when the margin between the top two similarities is small. Only the image encoder deploys; prototypes are computed offline. We evaluate four deployable encoders, each pinned to one public `open_clip` checkpoint: MobileCLIP2-S0/dfndr2b (11.4 M parameters), MobileCLIP-S1/datacompdr (21.5 M), MobileCLIP2-S2/dfndr2b (35.8 M) and MobileCLIP-B/datacompdr (86.3 M), with ViT-B-16-SigLIP2/webli (92.9 M) as a larger reference that is excluded from every quoted mean. Parameter counts are image-tower only.

For crops seen in training, the same frozen encoder with a linear probe reaches 82.2% over 166 fine-grained classes. A supervised convolutional network trained on the same classes reaches 89.3% but has no output unit for an unseen class, so its cross-crop accuracy is not low but undefined. This article concerns the unseen side.

3.3. Descriptor authoring strategies

Every strategy produces text for each class; the text is embedded by the frozen text tower and averaged into a prototype. The strategies differ only in how that text is written.

bare The class name alone, as “*disease on crop leaf*”, averaged over three prompt templates.

bare80 The same class name averaged over the 80 prompt templates of the original CLIP zero-shot recipe [15]. This tests whether a stronger template ensemble, rather than richer text, closes the gap.

crude The class name plus one generic keyword sentence (for example a stock description of rust or mildew).

rich The class name plus a hand-curated symptom paragraph retrieved from a 32-key keyword bank. Retrieval matches a disease name against the keys, so distinct diseases can receive identical text: at configuration C, 26 of the 51 held-out classes share their paragraph with another class.

grounded A symptom paragraph generated by an LLM under a prompt that requires every claim to cite a retrievable source, following the SAGE schema of pathogen, affected organs and visual symptoms (Section 3.4).

DCLIP Short visual-feature phrases per class, elicited with the question form of Menon and Vondrick [10], each embedded as “*class, which has feature*”; the class score is the mean similarity to its features.

CuPL Twelve full sentences per class describing what the disease looks like in a photograph, elicited with the question forms of Pratt et al. [14] and averaged into one prototype.

3.4. Descriptor generation

The **grounded** registry was generated by Claude Sonnet 4.6 at temperature 0 under a strict grounding prompt that forbids uncited claims. Because the generation endpoint cannot browse, raw citations are model-recalled; we web-verified the held-out diseases, replacing citations with sentences copied verbatim from reachable pages, and 23 of the 47 filled held-out records carry at least one page-verified field. The symptom text is what becomes the prototype; the citations provide the audit trail.

The **DCLIP** and **CuPL** registries were generated by an LLM in a chat interface, one crop per request, from the published question forms of each method with the category filled in. We added one instruction to both: describe the plant part the photograph actually shows. Several held-out classes are photographed as fruit or cereal heads rather than leaves (cigar end rot, for example, as a blackened banana tip), and a leaf description of those would describe something absent from the image. This instruction was informed by inspecting a few images, so both baselines received guidance the grounded registry did not; it strengthens them, which is the conservative

direction for the comparison. All 51 held-out classes were filled for both methods. Prompts, raw replies and the resulting registries are released.

3.5. Evaluation

We report top-1 accuracy over the held-out images of each configuration (micro-averaged), and summarise each strategy by its mean over the four deployable encoders. The three strategies that predate this study (bare, rich, grounded) were re-scored in the same run as the new ones, on the same cached image embeddings, and reproduce every previously released value exactly: 90 of 90 shared cells are bit-identical. Each descriptor registry is a single generation, so we read differences below the 2.4-point standard deviation measured between independently generated registries (Section 4.2) as within noise.

4. Results

4.1. Descriptor authoring is the dominant lever

Table 1 reports every strategy at every configuration on both label sets, and Figure 1 plots the full label set. Three patterns carry the argument.

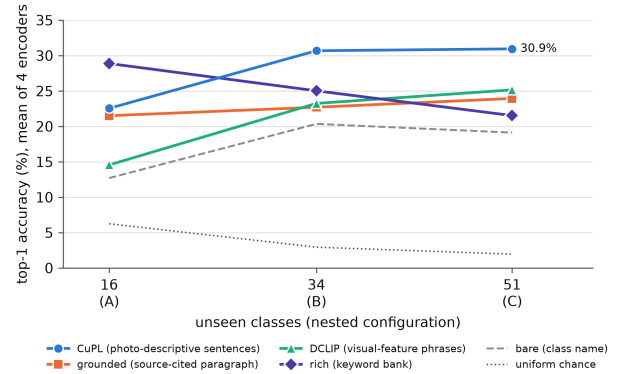


Figure 1: Mean unseen-crop top-1 over the four deployable encoders, by descriptor strategy, at the three nested label spaces (full label set). The keyword bank (rich) leads at 16 classes and falls as collisions grow; the per-class LLM methods do not collide, and CuPL leads at 34 and 51 classes. The dotted line is uniform chance.

Authoring outweighs model size Replacing a bare class name with the best descriptions gains 16.2, 10.3 and 11.8 points at configurations A, B and C. Moving from the 11.4 M to the 86.3 M encoder under the best descriptions gains 5.1, 4.0 and 4.7. Authoring therefore exceeds size by 2.5–3.1 times on the full label set. On the de-duplicated set the ratio is 1.2–2.8, the low end at configuration B, where the encoder spread widens to 10.0 points; authoring still exceeds size at every configuration on both label sets. The size gain is also the costlier of the two: it takes 5.8 times the on-device latency and 7.5 times the footprint (Section 4.4), whereas better descriptions are authored once, offline, and cost nothing per image.

Table 1

Unseen-crop top-1 accuracy (%) by descriptor strategy: mean over the four deployable encoders, at three nested label spaces (classes in parentheses), on the full and the de-duplicated label sets. The best strategy in each column is in bold. The authoring gain is the best strategy minus the class-name default; the encoder-size gain is the change from the 11.4 M to the 86.3 M encoder under the best strategy.

Strategy	Full label set			De-duplicated		
	A (16)	B (34)	C (51)	A (13)	B (26)	C (42)
bare	12.7	20.3	19.1	20.4	26.4	23.7
bare80	12.4	18.7	19.0	18.5	26.3	24.3
crude	14.1	18.7	16.7	24.5	26.0	21.7
rich	28.9	25.0	21.5	39.9	33.9	27.4
grounded	21.5	22.7	23.9	29.2	32.8	30.8
DCLIP	14.6	23.2	25.2	22.8	31.5	31.3
CuPL	22.5	30.7	30.9	33.1	38.5	37.5
Authoring gain (best – bare)	16.2	10.3	11.8	19.6	12.1	13.7
Encoder-size gain (86.3 – 11.4 M)	5.1	4.0	4.7	6.9	10.0	7.2

A stronger template ensemble does not substitute for richer text Averaging the class name over the 80 CLIP templates changes accuracy by -0.3 , -1.7 and -0.1 points at A, B and C relative to three templates. The gain from descriptions is therefore not an artefact of a weak class-name baseline; against the 80-template baseline it is unchanged or larger.

Which method wins depends on the label space At configuration A, with 16 classes, the keyword bank leads at 28.9%. Its prototypes collide, however, and the collisions grow with the label space, from 2 classes at A to 26 at C; its accuracy falls to 21.5% at C. The two per-class LLM methods cannot collide. CuPL leads at B and C with 30.7% and 30.9%, ahead of source-grounded text by 8.0 and 7.0 points and of the keyword bank by 5.7 and 9.4. Figure 2 shows the lead is not a property of one backbone: at B and C, CuPL beats source-grounded text on all four deployable encoders, by 4.8–10.1 points at B and 5.5–8.6 at C, and on the de-duplicated set it leads on all four encoders at every configuration. The margin is roughly three times the 2.4-point spread between independently generated registries, so a single generation does not account for it. DCLIP, by contrast, is level with source-grounded text at B and C ($+0.5$ and $+1.2$ points) and falls 7.0 points behind it at A.

At configuration C the best strategy reaches 30.9%, 7.4 times the 4.2% majority-class prior and 15.8 times uniform chance.

4.2. What separates the methods

Source-grounded text differs from CuPL in three respects that could each explain the gap: it is required to cite sources, it is embedded as one paragraph rather than as a sentence ensemble, and it is different text. Table 2 isolates each, holding the other two fixed.

Citation A matched control generated the same registry twice, with and without the requirement to cite a retrievable source, using the same model, schema and seven sampling

seeds. On the 41 classes both versions filled at every seed, the version with sources leads by $+1.96$ points (95% interval -0.23 to $+4.15$ over seeds); on all 51 classes, by $+0.94$ (-0.52 to $+2.39$). Both intervals include zero, and the paired figure is the optimistic of the two because its classes are those on which grounded generation succeeded. Requiring a source therefore costs no measurable accuracy. The seed-to-seed standard deviation, 2.4 points, is the registry-to-registry spread used throughout this article. We do not claim equivalence, which would need a margin fixed in advance.

Construction The source-grounded paragraph averages 199 tokens against the text encoder’s 77-token window, so 43 of the 51 held-out prototypes are truncated and most of each description never reaches the encoder. CuPL embeds twelve short sentences, none truncated. To test whether this explains the gap, we split each source-grounded paragraph into its sentences and embedded them as a CuPL-style ensemble, changing no words. Truncated prompts fall from 43 to 10. At configuration C accuracy moves by $+0.3$ points, recovering 4% of the gap to CuPL; on the de-duplicated set, $+0.9$ points and 13%. How the prototype is built explains almost none of the difference.

Text What remains, 6.7 points at configuration C on the full label set and 5.9 on the de-duplicated set, lies in the text. The source-grounded registry follows the SAGE schema of pathogen, affected organs and symptoms, written in the register of a pathology reference; CuPL’s sentences are written as descriptions of a photograph. CLIP’s text tower was trained on image captions, and the second register is closer to them. We name this as the likely explanation rather than a demonstrated one: the CuPL registry also differs from the grounded one in the model and interface that generated it, and a registry that is both source-grounded and photo-descriptive, which would separate register from provenance, is the natural next experiment.

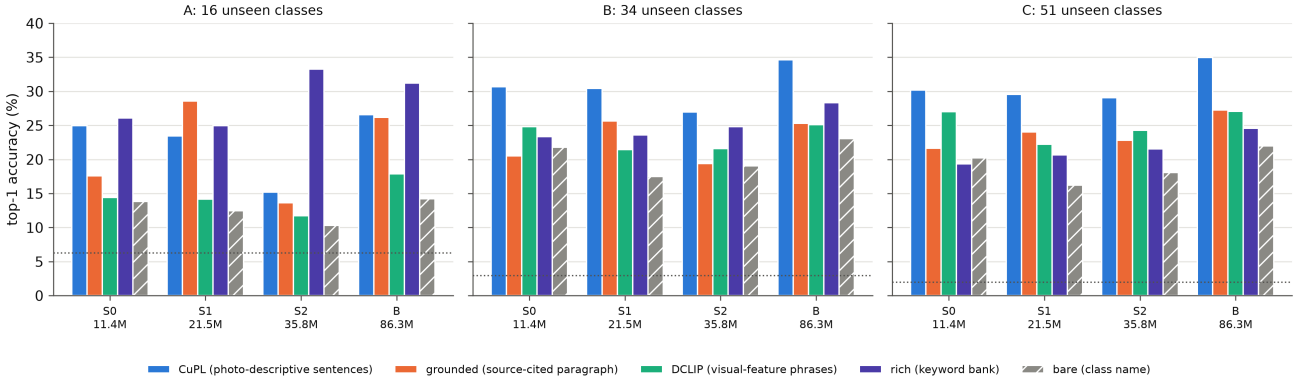


Figure 2: Unseen-crop top-1 by encoder and descriptor strategy at each configuration (full label set). At 34 and 51 classes CuPL is the tallest bar on every deployable encoder.

Table 2

Decomposition of the accuracy gap between source-grounded text and CuPL at configuration C (51 and 42 classes), in points of mean top-1 over the four deployable encoders. Each row changes one factor and holds the other fixed; construction and text sum to the total gap before rounding, so the rounded entries may differ from it by 0.1. The citation requirement is tested separately by a seeded control (Section 4.2): removing it changes accuracy by an amount whose 95% interval includes zero.

Factor	Comparison	Held fixed	Full	De-dup.
Construction	grounded → grounded, split into sentences	text	+0.3	+0.9
Text	grounded, split → CuPL	construction	+6.7	+5.9
Total gap	grounded → CuPL	—	+7.0	+6.7

4.3. Shortlisting and abstention

A field tool can offer a short list and decline when unsure, so top-1 is not the only relevant metric. Table 3 reports top-5 accuracy and selective prediction for the source-grounded registry and the keyword bank, the two strategies for which they were measured. On 51 classes the source-grounded shortlist contains the correct disease 59–75% of the time across the four deployable encoders, against 9.8% for a random five-item list; on 16 classes, 60–77% against 31.3%. The shortlist therefore earns most of its value in the larger label spaces. The margin-ranked gate orders confidence correctly: selective accuracy rises at every step of the reported coverage grid, from full coverage down to half in steps of ten, in all 30 encoder, strategy and configuration cells (Figure 3). Its effect is modest, about two points of top-1 for declining one image in five; its value is that the system can decline rather than guess. Top-5 and abstention were not measured for CuPL, whose higher top-1 suggests but does not establish a higher top-5.

4.4. On-device cost

Table 4 reports the deployable image encoders on a laptop processor (AMD Ryzen 7 7840HS, 16 threads, ONNX Runtime 1.26.0, batch 1, 224-pixel input, median of 50 runs). The 11.4 M encoder runs at 17.4 ms per image, about 58 images per second, in 45.8 MB, and ONNX Runtime is about 2.7 times faster than eager PyTorch. Figure 4 places the encoders on the accuracy–latency plane under CuPL descriptions.

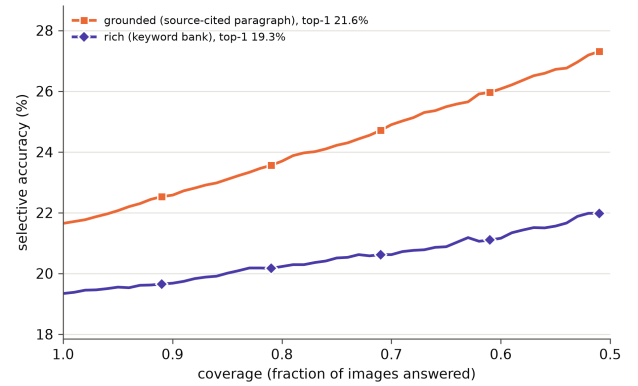


Figure 3: Selective accuracy against coverage for the margin-ranked gate at 51 classes, on the smallest encoder. Accuracy rises at every step of the reported grid, from full coverage to half; below half coverage the surviving sample is too small to estimate accuracy stably and is not plotted.

INT8 depends on architecture A default dynamic-quantisation call makes the three hybrid convolution–transformer encoders 18–19 times *slower*. A static quantise–dequantise pipeline with per-channel weights and calibrated activations recovers 5.1–9.4 times of that loss but still does not beat FP32 on those encoders. The pure-transformer encoder behaves oppositely and becomes 2.2 times faster. The quantised graphs show why: the quantiser leaves 119, 215 and 235 convolutions unconverted

Table 3

Top-5 and selective prediction, source-grounded against the hand-curated bank. acc@cov80 = accuracy when the 80% most confident predictions are kept; AURC = area under the risk-coverage curve, lower is better.

Config	Classes	Model	grounded				rich			
			Top-1	Top-5	AURC ↓	acc@cov80	Top-1	Top-5	AURC ↓	acc@cov80
A	16	MobileCLIP2-S0	17.6%	61.9%	0.6986	19.4%	26.1%	59.6%	0.6776	28.9%
A	16	MobileCLIP-S1	28.6%	60.3%	0.5015	33.0%	24.9%	55.3%	0.6287	28.6%
A	16	MobileCLIP2-S2	13.7%	61.2%	0.7426	14.8%	33.3%	63.6%	0.5840	37.6%
A	16	MobileCLIP-B	26.2%	77.1%	0.6141	28.4%	31.2%	69.9%	0.5243	35.0%
A	16	ViT-B-16-SigLIP2	26.8%	76.9%	0.6532	28.5%	30.4%	68.5%	0.6197	33.4%
B	34	MobileCLIP2-S0	20.5%	70.0%	0.7077	22.3%	23.3%	63.3%	0.6775	25.1%
B	34	MobileCLIP-S1	25.6%	63.7%	0.5944	29.2%	23.6%	61.5%	0.6018	26.6%
B	34	MobileCLIP2-S2	19.4%	71.6%	0.7128	20.3%	24.8%	66.4%	0.6114	27.1%
B	34	MobileCLIP-B	25.3%	76.0%	0.5941	27.4%	28.4%	67.5%	0.5450	30.5%
B	34	ViT-B-16-SigLIP2	27.3%	74.7%	0.6623	28.3%	29.9%	72.4%	0.5721	32.4%
C	51	MobileCLIP2-S0	21.6%	62.9%	0.7168	23.7%	19.3%	61.7%	0.7748	20.2%
C	51	MobileCLIP-S1	24.0%	59.0%	0.6430	26.9%	20.7%	58.1%	0.7008	22.6%
C	51	MobileCLIP2-S2	22.8%	68.3%	0.7356	23.6%	21.5%	66.1%	0.6970	23.4%
C	51	MobileCLIP-B	27.3%	74.6%	0.6169	29.7%	24.6%	66.0%	0.6548	25.8%
C	51	ViT-B-16-SigLIP2	28.4%	67.7%	0.6155	30.7%	23.1%	67.1%	0.6813	24.9%

Confidence is the top-1 minus top-2 similarity margin. Selective accuracy rises at every step of the reported coverage grid (100% down to 50% in tens) in all 30 cells, confirming the signal is correctly ordered; the gate itself buys about two points of top-1 for refusing one image in five.

Table 4

On-device cost of the deployable image encoder (CPU, batch 1, 224×224). The four latency columns are milliseconds; the final two are megabytes.

Model	Params (M)	Torch FP32 ↓	ONNX FP32 ↓	INT8 dyn. ↓	INT8 static ↓	FP32 ↓	INT8 ↓
MobileCLIP2-S0	11.4	47.1	17.4	320.0	62.2	45.81	12.92
MobileCLIP-S1	21.5	99.8	33.7	614.9	79.5	86.51	24.29
MobileCLIP2-S2	35.8	119.0	49.2	923.8	98.7	143.62	39.16
MobileCLIP-B	86.3	130.7	100.8	60.4	46.4	345.55	87.4

Latency in ms (median of 50 runs), ONNX Runtime 1.26.0. INT8 makes the three hybrid conv-transformer tiers *slower* while shrinking them $\sim 3.5\times$; the one pure-transformer model here gets faster instead. The transformer side of that contrast rests on a single row.

in the three hybrids against 3 in the transformer, so on the hybrids the runtime must dequantise, run a floating-point convolution and requantise hundreds of times per inference. For hybrid encoders on an x86 CPU runtime, INT8 is therefore a size lever, shrinking the smallest encoder 3.5 times to 12.9 MB, not a speed lever. The contrast rests on one transformer row, which is also the largest model in the table, so architecture and scale are not separated; the convolution counts are what make architecture the likelier cause, and Arm processors, with well-optimised INT8 depthwise kernels, may reverse the hybrid result.

5. Discussion

Spend effort on descriptions For a frozen compact encoder, the text prototypes decide whether cross-crop diagnosis works. Across a 7.6-fold range of encoder size, the best descriptions buy 2.5–3.1 times more accuracy than the largest encoder does, and they are paid for once, offline, whereas encoder size is paid for on every image in latency and memory.

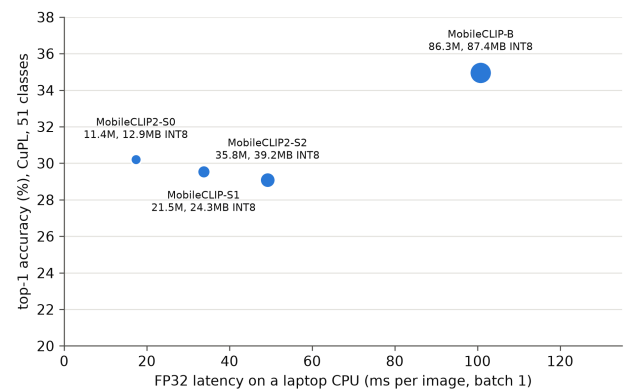


Figure 4: Unseen-crop top-1 under CuPL descriptions at 51 classes against single-image FP32 latency on a laptop CPU. Marker area is proportional to parameter count; labels give the INT8 footprint. The smallest encoder minimises latency and footprint; the largest gives the highest accuracy at 5.8 times the latency.

The practical recommendation follows: deploy the smallest encoder that meets the latency budget, and invest in the descriptions.

Write for the camera Among the methods tested, descriptions written as accounts of a photograph worked best, and the decomposition of Section 4.2 locates the advantage in the text rather than in how it is embedded or in whether it cites sources. Source-grounded text written in the register of a pathology reference, with its pathogen and taxonomy fields, lags by about seven points. The implication for authoring agricultural descriptors is concrete: describe colour, shape, texture, margin and distribution as a camera would record them, and leave taxonomy out of the prototype.

Auditability is free Requiring every description to cite a retrievable source did not measurably cost accuracy. A deployment that must justify its outputs, as an agricultural advisory service may, can therefore demand source-grounded text without an accuracy penalty from the citation requirement itself. What it should not assume is that sourcing improves accuracy; it does not, and the register in which the grounded text is written currently costs it seven points. A registry that is both cited and photo-descriptive would, if the register explanation holds, combine the audit trail with the best accuracy.

What does not work Two alternatives to a frozen compact encoder with good descriptions are ruled out. A supervised convolutional network is the stronger choice for a fixed set of known crops, reaching 89.3% on 166 seen classes against 82.2% for the frozen probe, but it cannot represent an unseen crop at all. Fine-tuning the encoder on the seen crops raises seen accuracy but lowers unseen accuracy, from 21.6% to 16.4% on the smallest encoder at configuration C, so specialisation trades away the transfer this system exists to provide; we keep the encoder frozen.

Absolute accuracy Fine-grained cross-crop top-1 remains modest: on 51 unseen classes the four description-based strategies reach 21.5–30.9%. We therefore frame the system as a shortlist with the ability to abstain rather than as an automatic diagnostician, and cross-crop transfer as a problem this work advances at edge scale rather than solves.

6. Limitations

One generation per registry Each descriptor registry is a single generation. The 2.4-point registry-to-registry spread bounds how far any single cell could move, and the central margins here, CuPL over source-grounded text at 7.0 points and authoring over size by 2.5–3.1 times, are well outside it. Smaller differences, such as DCLIP against source-grounded text, are not. Regenerating the DCLIP and CuPL registries over further seeds would turn their directions into measured effects.

Register is confounded with provenance The source-grounded and CuPL registries differ in register and also in the model and interface that produced them, and the CuPL registry was generated one crop per request, so the model saw sibling classes. We attribute the gap to the text, which the decomposition supports, but name register as its likely cause rather than a demonstrated one.

The baselines received image-informed guidance The instruction to describe the photographed plant part was informed by inspecting a few images. It favours DCLIP and CuPL, the conservative direction for a comparison that CuPL wins, but the source-grounded registry did not receive it.

Shortlist and abstention were measured on two strategies Top-5 and selective prediction are reported for the source-grounded registry and the keyword bank only, not for CuPL.

The nested design does not isolate label-space size Each configuration adds new crops as well as classes, and at configuration A the held-out crops are easier than the seen pool on a common pool of 78 classes (19.0% against 9.0% under the keyword bank). Scoring one fixed image set against label spaces of increasing size would separate the two effects.

Head selection is unsolved The system diagnoses unseen crops assuming the operator knows whether a crop is one the probe was trained on. Nothing in the pipeline infers which head applies; an open-set gate [16] with its own error rate, and the resulting end-to-end accuracy, is the experiment that would make these components a fielded system.

Scope Results cover 18 crops of one dataset at a pinned revision, on one evaluation machine. Source verification is partial: 23 of the 47 filled held-out grounded records carry a page-verified field, and the remainder carry model-recalled citations, released for independent checking.

7. Conclusion

A frozen compact vision–language model can diagnose plant diseases on crops it was never trained on, at 17.4 ms per image on a laptop CPU, and how its descriptions are written decides how well. Across seven authoring strategies, three label spaces and four encoders, the best descriptions buy more accuracy than the largest encoder, at no per-image cost. Photo-descriptive per-class sentences work best, reaching 30.9% on 51 unseen classes, 7.4 times the majority-class prior. Requiring descriptions to cite sources costs nothing measurable, and the remaining gap between cited and photo-descriptive text lies in the text rather than in citation or embedding. The lever for edge deployment is the descriptions, not the parameters.

CRedit authorship contribution statement

Pinni Venkata Abhiram: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation,

Writing – original draft, Writing – review and editing, Visualization, Project administration. **Gaurav Shrivastava:** Methodology, Software, Validation, Investigation, Writing – review and editing. **Rahul Ananthasayanam:** Validation, Investigation, Writing – review and editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

All data underlying this study are public. The images are the SAGE crop-disease dataset [2] at revision bc9bd2899f, released under the MIT licence and used without modification other than the crop and per-class subsetting and the 288-pixel resizing described in Section 3.1. The descriptor registries for every strategy, including the DCLIP and CuPL prompts and raw replies, every result file, the table and figure generators, and the full experimental code are released at <https://doi.org/PENDING-ZENODO-DOI>, which reproduces every number and figure in this article from the released result files.

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

Two distinct uses are declared. First, as a methodological component: large language models generate the descriptor registries that are the object of study, the source-grounded registry through an API (Section 3.4) and the DCLIP and CuPL registries through a chat interface. The generated text is released with the data so that it can be inspected. Second, in preparing the manuscript: the authors used a large language model to assist with drafting and editing prose and with writing the analysis scripts that produce the tables and figures from the released result files. No text was accepted without review, every reported number is computed from a released result file rather than transcribed, and the authors reviewed and edited all content and take full responsibility for the publication.

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